

SHUBHAM KUSHWAH

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SUMMARY

Entry-level Backend Developer specializing in Java and Spring Boot. Built and containerized distributed REST APIs on Oracle Cloud (OCI), achieving a peak throughput of 9,317 RPS by replacing traditional database locks with atomic Redis Lua scripting. Possesses strong CS fundamentals (OS, DBMS, Computer Networks) and demonstrated problem-solving skills with 250+ LeetCode problems solved (1650 Rating, Top 20 % globally). Seeking an SDE-1 backend role.

TECHNICAL SKILLS

Languages: Java 17/21, SQL

Frameworks: Spring Boot, Spring Security, Spring Data JPA, Hibernate

Databases & Caching: MySQL (ACID, Indexing), PostgreSQL, Redis (Lua Scripting)

DevOps & Infrastructure: Docker, Oracle Cloud (OCI), Nginx Reverse Proxy, Git, Postman

Testing & Benchmarking: k6 Load Testing, Concurrency Stress Testing

Core Concepts: Database Locking (Optimistic/Pessimistic), RESTful API Design, JWT Authentication, OOP Distributed Systems, Cloud Architecture

TECH PROJECTS

Flash Sale Concurrency Engine | Java 21, Spring Boot, Redis, MySQL, Docker, OCI, Nginx [GitHub](#) | [Live Portfolio](#)

- System Overview:** Engineered a distributed backend system to mimic high-traffic e-commerce flash sales, processing massive surges of checkout requests with a strict zero-oversell guarantee.
- Cloud Architecture:** Containerized the application via Docker and deployed on Oracle Cloud (OCI). Initially scaled across multiple App Nodes behind a Load Balancer for stress testing, currently consolidated behind an Nginx reverse proxy.
- Concurrency Optimization:** Replaced standard MySQL pessimistic row-locking (which choked connection pools at 460 req/sec) with an atomic **Redis Lua caching layer** to completely eliminate database lock contention.
- Peak Performance:** Achieved a peak throughput of **9,317 requests per second (RPS)** with a sub-second p(95) latency of **754ms** by decoupling high-speed cache decrements from asynchronous DB persistence.
- Extreme Scale Testing:** Validated system resilience using dedicated **k6 distributed load runners**, sustaining **100,000 Concurrent Virtual Users (VUs)** while tuning Linux OS socket limits (`ulimit -n 650000`) to prevent TCP port exhaustion.

Trading Journal API | Java 21, Spring Boot, Spring Security, JWT, MySQL [GitHub](#) | [Live Interactive API](#)

- Developed a RESTful trade analytics platform with a stateless security architecture using `OncePerRequestFilter` and JSON Web Tokens.
- Enforced strict cross-user data isolation by extracting authenticated security contexts directly within the service layer, preventing unauthorized resource access.
- Leveraged Java Streams API (`groupBy`, `mapToDouble`) to aggregate complex portfolio statistics including real-time P&L and strategy win-rates.
- Optimized high-volume trade retrieval by implementing Spring Data `Pageable` and dynamic sorting for paginated database queries.
- Integrated Apache Commons CSV and multipart file handling to support programmatic data exports and trade screenshot storage.

EDUCATION

Institute of Technology and Management, Gwalior
B.Tech in Computer Science Engineering (CGPA: 7.01)

Madhya Pradesh
2021 – 2025

ACHIEVEMENTS

- Competitive Programming:** Achieved a maximum contest rating of 1650 on LeetCode (Top 20% Globally).
- Problem Solving:** Solved 250+ Data Structures and Algorithms problems, specializing in Java programming and execution efficiency.